

Effects of Heterogeneous Customers Schedule and Time in Queue Systems

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Outline

- ▶ Overview
- ▶ Objectives
- ▶ Optimal and Heuristic methods
- ▶ Validation
- ▶ Homogeneous Customers Schedule
- ▶ Heterogeneous Customers Schedule
- ▶ Conclusions

Overview

Queueing theory is the mathematical study of waiting lines, or queues. It plays an important role in modeling real life problems involving congestion in wide areas of applied sciences.

Some examples are:

- ▶ banks/supermarkets - waiting for a service
- ▶ public transport - waiting for a train
- ▶ computers - waiting for a response
- ▶ failure situations - waiting for a failure to occur e.g. in a piece of machinery or a bus

Overview (cont.)

We study a multi-hour queue problem, in which N customer classes have to visit M location classes in T discrete time slices.

- ▶ Each customer class has an average demand towards a location class.
- ▶ Each location class has an average service rate.
- ▶ At each time step, each location experiences an average arrival rate (work rate).

Objectives

- ▶ Develop the Optimal and an Heuristic Solutions, with AMPL and Python respectively, to minimize the overall delay in the system. As a result, each customer class will know its best schedule.
- ▶ Study how the delay in the system is affected by the time T .
- ▶ Study how the delay in the system is affected by adding heterogeneity in the customers schedule.



Figure: “Which task should I do first in order to minimize the waiting time?”



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Optimal and Heuristic methods

Both methods are based on the following setting. The standard notation is:

M/M/M/C_j/∞/FCFS

- ▶ M: inter-arrival time is random and described through a Poisson process.
- ▶ M: service-time distribution is exponential.
- ▶ M: number of servers, locations available
- ▶ $\bar{C}$: maximum number of jobs available in the system at all locations.
- ▶ ∞ : any customer class is accepted from the system.
- ▶ FCFS: first come first served.

In addition, our model is multi-hour. So the processes has its own time-varying structure.

Optimal and Heuristic methods (Cont.)

To minimize the overall delay in the system, we minimize the *load sensitive link delay* function:

$$\min F = \sum_{j,t} F_{jt} = \sum_{j,t} \frac{y_{jt}}{C_j - y_{jt}} \quad (1)$$

where C_j is the average service rate at location j , y_{jt} is the average work rate at location j at time t , F_{jt} is the delay in location j at time t and F is the total delay in the network.

Optimal and Heuristic methods

Optimal method: AMPL pseudo-code to minimize the overall average delay in the system F . The system is studied under a piece-wise linear approximation. The optimal method models cooperative costumers: all customers agree to change their schedule according to the overall minimum delay.

indices:

$i = 1, \dots, N$ customer classes
 $j = 1, \dots, M$ queues/locations
 $t = 1, \dots, T$ traffic hours

constants:

$h_{i,j}$ average demand rate from a specific customer in a specific location
 C_j average service rate in a specific location

variables:

u_{ijt} binary variable equal to 1 if customer i is in queue j at time t , otherwise 0
 y_{jt} average work rate in a specific location in a specific time
 r_{jt} link delay

objective:

minimize $F = \sum_{j,t} r_{jt}$

constraints:

$\sum_t u_{ijt} = 1$	$i = 1, \dots, N \quad j = 1, \dots, M$
$\sum_i u_{ijt} h_{ij} = y_{jt}$	$j = 1, \dots, M \quad t = 1, \dots, T$
$r_{jt} \geq y_{jt}$	$j = 1, \dots, M \quad t = 1, \dots, T$
$r_{jt} \geq 3y_{jt} - 2C_j/3$	$j = 1, \dots, M \quad t = 1, \dots, T$
$r_{jt} \geq 10y_{jt} - 16C_j/3$	$j = 1, \dots, M \quad t = 1, \dots, T$
$r_{jt} \geq 70y_{jt} - 178C_j/3$	$j = 1, \dots, M \quad t = 1, \dots, T$
$r_{jt} \geq 500y_{jt} - 1468C_j/3$	$j = 1, \dots, M \quad t = 1, \dots, T$
$r_{jt} \geq 5000y_{jt} - 16318C_j/3$	$j = 1, \dots, M \quad t = 1, \dots, T$



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Optimal and Heuristic methods

Heuristic method: Python pseudo-code which selects randomly a customer to minimize its waiting times. After having selected all customers, the total average delay in the system is computed. The process is iterated *several times* due to stochasticity (randomly choose a customer). This heuristic method models egoistic costumers: each customer want to minimize his own delay.

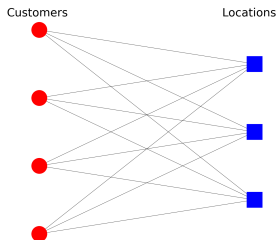
```
Data:  $T$  time windows
Queues  $\leftarrow$  a set of  $M$  locations with given service rates  $C$ 
Customers  $\leftarrow$  a set of  $N$  customers with given demands  $h$  to specific
locations
 $y \leftarrow$  average amount of work rate in a specific location at a specific
time
delay  $\leftarrow$  delay in a specific location at a specific time
Customers.done  $\leftarrow$  an empty set which will contain all customers
with their best schedule
Initialization: Set  $T$ ,  $M$  and  $N$ ;
Set  $h$  elements (dimensions  $N \times M$ );
Set  $C$  elements (dimension  $M$ );
Set  $y$  elements to zero (dimension  $M \times T$ );
Set delay elements to zero (dimension  $M \times T$ );
for Several times do
    while Customers.done not includes all the Customers available
    do
        Choose randomly a customer class not in Customers.done;
        Update delay in the queues. Each element will be equal to  $F_{jt}$ 
        computed from eq. 1;
        Compute customer's schedule to minimize his delay.;
        if A schedule is possible then
            | The customer will choose a time to visit a specific location
            | able to minimize his delay.
        else
            | The customer will have to wait that choose a time to visit
            | a specific location able to minimize his delay.
        end
        Append this customer class to Customers.done;
    end
end
```



Validation

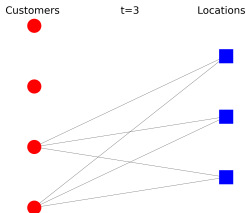
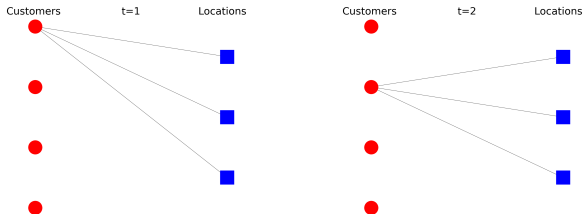
We study a small system of $N = 4$ customer classes and $M = 3$ location classes. We represent the system as a bipartite network in which red circles represent customers and blue squares locations. A link is present if a specific customer class have to visit a specific location. The width of the link represents the value of the demand. In the validation setting, all the weight are equivalent. We also assume that all locations have the same average service rate.

The goal is to find at what instant of time each customer should visit a specific location in order to minimize the average delay in the network.



Validation (cont.)

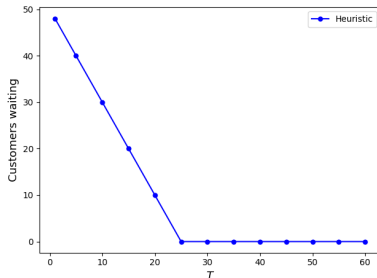
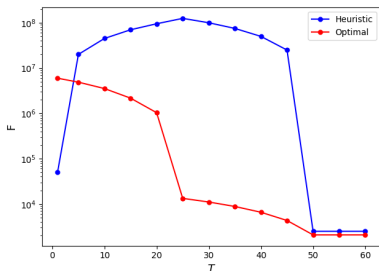
We set $T = 3$ and the solution from AMPL is given as follow.



Homogeneous Customers Schedule

First, we compare the Optimal and Heuristic solutions, indicated generally with F as a function of T (Figure on the left). Second, we assess how many customer classes are waiting as a function of T (Figure on the right).

We set $N = 50$, $M = 50$ and study the delay as a function of T . All the links have the same weights/demands (equal to $\bar{h} = 0.5$) and all the locations have the same service rates (equal to $\bar{C} = 1.0001$). We iterate 50 times the Heuristic algorithm and plot the 95% confidence interval. Since all customers are equivalent, all independent iterations present the same results.



Homogeneous Customers Schedule (Cont.)

As expected, we observe that the optimal total average delay is generally lower than the heuristic total average delay. The exception encountered at $T = 1$ can be explained by looking at the number of customers waiting. If $0 < T < 25$, the number of customers waiting diminished and the heuristic total average delay F grows. This effect is a consequence of our algorithm implementation: prior to put a customer in the waiting line, he chooses one queue at a certain time and this queue will be full for that particular time. Therefore, a single queue could be full for T times and its total delay is proportional to T .

If $25 \leq T < 50$, the queues can accomodate all the customers and the heuristic solution decays as T increases. Additionally, the optimal solution becomes much cheaper than the heuristic: the AMPL objective function is subject to less penalties.

If $T > 50$, the two delays do not vary anymore. The time windows T are enough to already accomodate the customer's demands.



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Homogeneous Customers Schedule (Cont.)

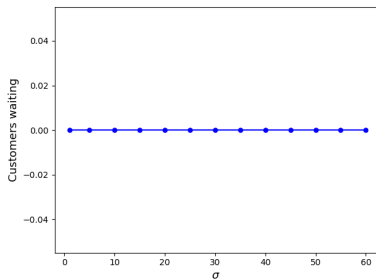
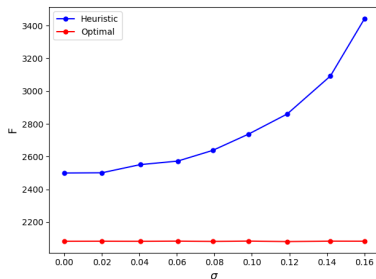
We define $T_{th,1} = 25$ and $T_{th,2} = 50$ in order to separate the three regimes described in the previous slide.

Obviously, $T_{th,1} = T_{th,1}(M, N, \bar{h}, \bar{C})$ and $T_{th,2} = T_{th,2}(M, N, \bar{h}, \bar{C})$ even if a more detailed studied is needed to assess the exact relation. Indeed, we must notice that, given $M/N = 1$, it is not surprising that $T_{th,1}/T_{th,2} \approx \bar{h}/\bar{C}$.

Heterogeneous Customers Schedule

Previously, all customers had the same demands/link weights equal to $\bar{h} = 0.5$. At this point, we generate a set of demands distributed as a Truncated Gaussian distribution (using a built-in function in python to constrain the demands between 0 and $\bar{C} = 1$). We set the mean value equal to $\bar{h} = 0.5$ and we study the total delay in the system as a function of the standard deviation σ .

We set $T = T_{th,2} = 50$ and all the other parameters follows from above. We observe that the optimal solution is not affected from the presence of heterogeneous customers schedule. Whilst, the heuristic solution increases as the heterogeneity in the system increases. The number of customers is not affected by the presence of heterogeneity.



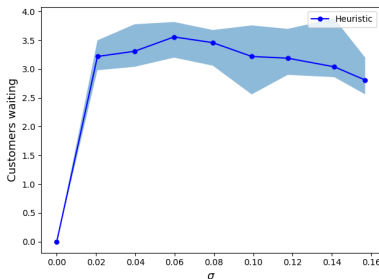
Heterogeneous Customers Schedule (Cont.)

At this point, we focus our discussion on the waiting lines and on the heuristic solution. All the parameters are unchanged but the time windows: $T_{th,1} < T = 35 < T_{th,2}$.

Interestingly, the number of customers waiting jumps as soon a small heterogeneity is added to the system. However, its value slightly decreases while the overall heterogeneity increases. A reason for this unexpected phenomenon could be attributed to the choice $2\bar{h} + \epsilon = \bar{C}$ and $\epsilon \ll 1$. It implies that 2 customers perfectly fit in a service.

Therefore, if either one of the two customers is just slightly greater, they do not fit anymore and one of them has to move in the waiting list.

95% confidence interval is plotted after 50 independent iterations.

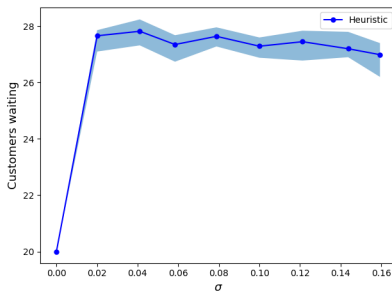


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Heterogeneous Customers Schedule (Cont.)

A similar argument applies for $0 < T = 15 < T_{th,1}$.



Conclusions

In this project, we present an AMPL and a Python code to respectively find the optimal solution and an heuristic solution of a multi-hour queue problem. The optimal method models cooperative behavior. The heuristic method egoistic individuals. The main findings are:

- ▶ If a set of individuals cooperates, the overall delay in the system is not affected by the presence of an heterogeneous customers schedule.
- ▶ If individuals are egoistic and want just to minimize their own delay, the overall delay in the system increases as the customers schedule become more heterogeneous.

Conclusions

Furthermore, we find that two system-dependent critical times exist:

- ▶ $T_{th,1} = T_{th,1}(M, N, \bar{h}, \bar{C})$ determines a threshold time above which some customers have to join the waiting line.
- ▶ $T_{th,2} = T_{th,2}(M, N, \bar{h}, \bar{C})$ determine a second threshold time above which the total delay in the system does not change.

A further study should be carried on to assess the exact dependence of $T_{th,1}$ and $T_{th,2}$ on the system parameters $M, N, \bar{h}$ and $\bar{C}$.